SZABIST
University

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ROBOTICS & ARTIFICIAL INTELLIGENCE DEPARTMENT

Total Marks:_____

Obtained Marks: _____

Computing Vision

Lab Task # 07

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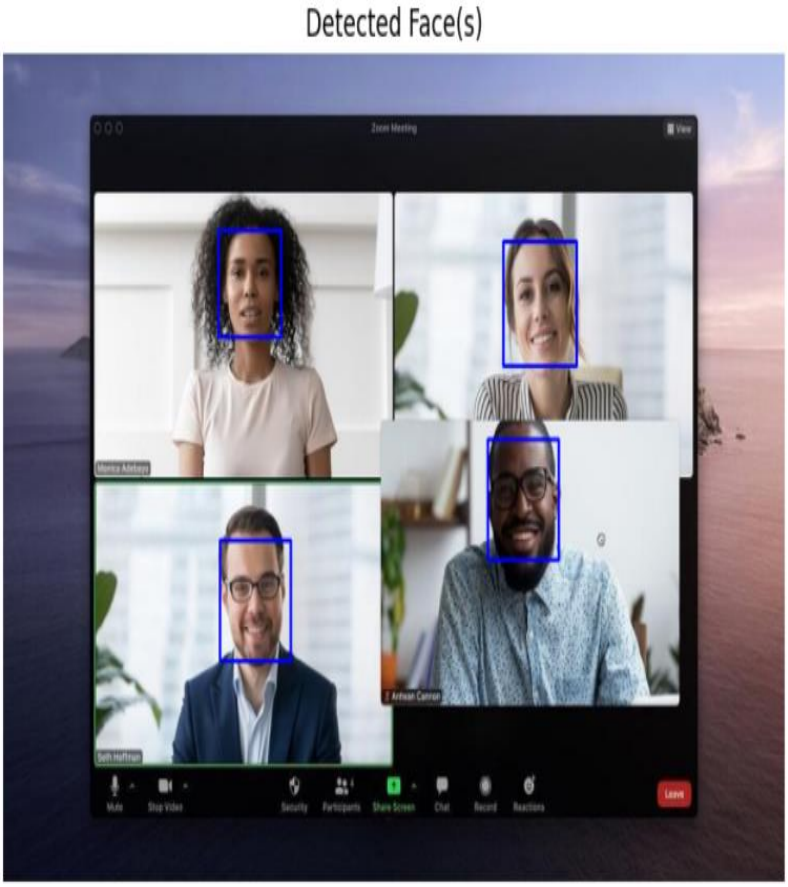
Code Example 1: Face Detection Using Haar Cascade

Security cameras in offices, banks, and public spaces use face detection as a first step before applying facial recognition or person tracking. Detecting faces helps trigger alerts or log entries when a person is identified in the camera's field of view.

```python
# Code Example 1: Face Detection Using Haar Cascade
import cv2
import matplotlib.pyplot as plt
face_cascade = cv2.CascadeClassifier(cv2.data.haarcascades + 'haarcascade_frontalface_default.xml')
img = cv2.imread("D:/23108106/A22.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
faces = face_cascade.detectMultiScale(gray, scaleFactor=1.1, minNeighbors=5)
for (x, y, w, h) in faces:
    cv2.rectangle(img, (x, y), (x + w, y + h), (255, 0, 0), 2)
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
plt.figure(figsize=(8, 6))
plt.imshow(img_rgb)
plt.title("Detected Face(s)")
plt.axis('off')
plt.show()
```



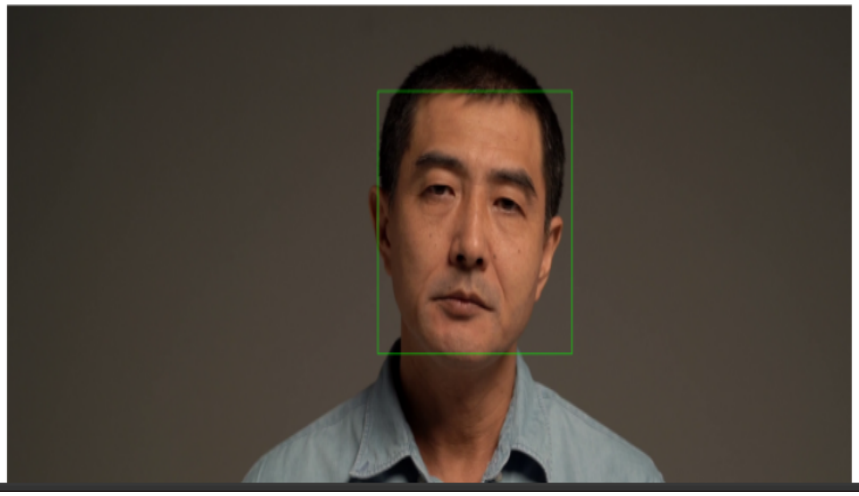
Code Example 2: Real-Time Face Detection with Webcam

A local security agency is developing a smart surveillance system for monitoring restricted areas such as laboratories, bank vaults, or examination halls. To reduce manual monitoring efforts, the system needs to automatically detect human presence by identifying faces in real-time from CCTV footage. The goal is to detect intrusions, unauthorized entries, or simply log face-based entries in video archives.

You are part of the development team and assigned the task of building the face detection module that can process a short video clip, detect human faces, and highlight them frame by frame.

```python
# Code Example 2: Real-Time Face Detection with Webcam
import cv2
import matplotlib.pyplot as plt
face_cascade = cv2.CascadeClassifier(cv2.data.haarcascades + 'haarcascade_frontalface_default.xml')
cap = cv2.VideoCapture("D:/23108106/122389-725067792.mp4")
frame_count = 0
while True:
    ret, frame = cap.read()
    if not ret or frame_count >= 100:
        break
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    faces = face_cascade.detectMultiScale(gray, 1.1, 4)
    for (x, y, w, h) in faces:
        cv2.rectangle(frame, (x, y), (x + w, y + h), (0, 255, 0), 2)
    rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
    if frame_count % 20 == 0:
        plt.figure(figsize=(8, 5))
        plt.imshow(rgb)
        plt.title(f"Frame {frame_count} with Detected Face(s)")
        plt.axis('off')
        plt.show()
    frame_count += 1
cap.release()
```

Frame 0 with Detected Face(s)



Case Study 1: Automated Celebrity Face Recognition for Media Archiving

A media company is working on a digital archiving solution for its vast image database of public figures and celebrities. The goal is to automatically tag and organize images based on the identities of individuals in them. Your role as a machine learning engineer is to create a system that can train on a labeled dataset of known faces (e.g., actors or public figures), and then use the trained model to recognize and label faces in new images.

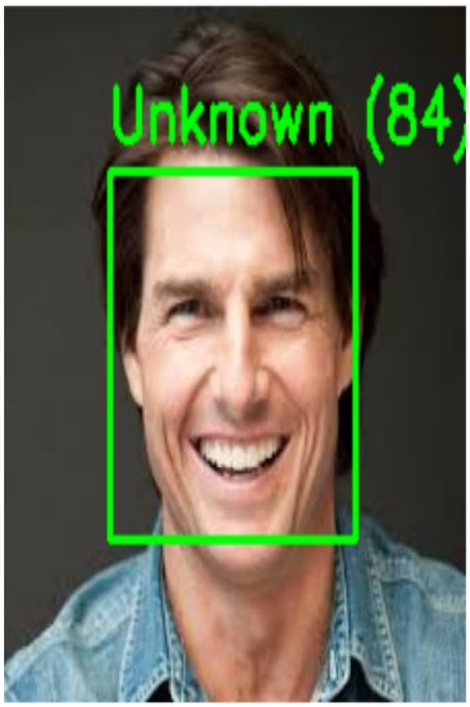
You have been provided with a dataset of facial images grouped in folders by person's name. The system should train a recognizer and then test its prediction ability on a new image.

```python
# Case Study 1: Automated Celebrity Face Recognition for Media Archiving
import os
import cv2
import numpy as np
from matplotlib import pyplot as plt
def load_dataset(dataset_path):
    images = []
    labels = []
    label_dict = {}
    current_label = 0
    for person_name in os.listdir(dataset_path):
        person_path = os.path.join(dataset_path, person_name)
        if os.path.isdir(person_path):
            label_dict[current_label] = person_name
            for img_name in os.listdir(person_path):
                img_path = os.path.join(person_path, img_name)
                img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)
                if img is not None and img.shape[0] > 0 and img.shape[1] > 0:
                    images.append(img)
                    labels.append(current_label)
                else:
                    print(f"Failed to load or invalid image: {img_path}")
            current_label += 1
    return images, labels, label_dict
dataset_path = "D:/23108106/Dataset/Faces"
images, labels, label_dict = load_dataset(dataset_path)
recognizer = cv2.face.LBPHFaceRecognizer_create()
recognizer.train(images, np.array(labels))
recognizer.save('D:/23108106/face_recognizer.yml')
with open("label_map.txt", "w") as f:
    for label, name in label_dict.items():
        f.write(f"{label}:{name}\n")
recognizer.read('D:/23108106/face_recognizer.yml')
label_map = {}
with open("label_map.txt", "r") as f:
    for line in f:
        label, name = line.strip().split(":")
        label_map[int(label)] = name
face_cascade = cv2.CascadeClassifier(cv2.data.haarcascades + 'haarcascade_frontalface_default.xml')
test_img_path = "D:/23108106/A23.jpg"
```

```
img = cv2.imread(test_img_path)
if img is None:
    raise ValueError("Test image not loaded! Check the path.")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
faces = face_cascade.detectMultiScale(
    gray,
    scaleFactor=1.1,
    minNeighbors=3,
    minSize=(30, 30)
)
confidence_threshold = 80
if len(faces) == 0:
    print("No faces detected!")
else:
    for (x, y, w, h) in faces:
        roi_gray = gray[y:y+h, x:x+w]
        label, confidence = recognizer.predict(roi_gray)
        name = label_map.get(label, "Unknown")
        display_name = name if confidence < confidence_threshold else "Unknown"
        cv2.rectangle(img, (x, y), (x+w, y+h), (0, 255, 0), 2)
        cv2.putText(img, f"{display_name} ({int(confidence)})",
                    (x, y-10), cv2.FONT_HERSHEY_SIMPLEX, 0.8, (0, 255, 0), 2)
        if display_name != "Unknown":
            print(f"Unlocked: {display_name} (Confidence: {int(confidence)})")
        else:
            print(f"Access Denied (Confidence: {int(confidence)})")
plt.figure(figsize=(8,5))
```

```
plt.figure(figsize=(8,5))
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.axis('off')
plt.show()
```

```
Access Denied (Confidence: 84)
```



Case Study 2: Face Unlock in Smartphones

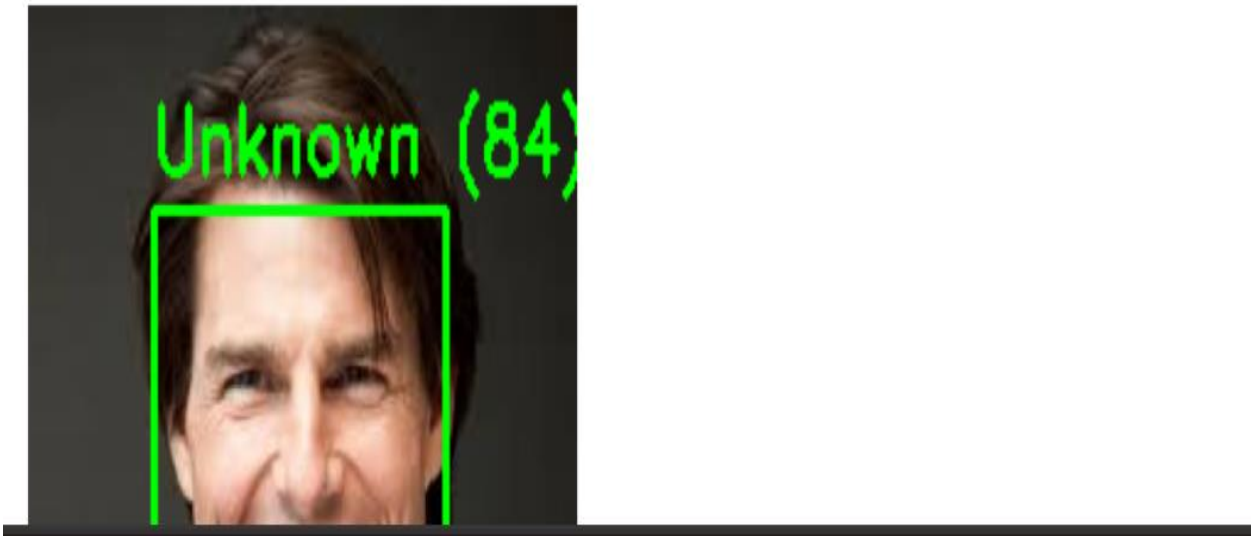
Smartphones use front cameras to unlock devices using the owner's face. The phone stores the owner's face pattern securely. Every time the device is accessed, it captures a new image, compares it with the stored one, and unlocks if matched.

Goal: Simulate face unlocking by training on one user's face and testing with another image to verify identity.

```
# Case Study 2: Face Unlock in Smartphones
import os
import cv2
import numpy as np
from matplotlib import pyplot as plt
def load_dataset(dataset_path):
    images = []
    labels = []
    label_dict = {}
    current_label = 0
    for person_name in os.listdir(dataset_path):
        person_path = os.path.join(dataset_path, person_name)
        if os.path.isdir(person_path):
            label_dict[current_label] = person_name
            for img_name in os.listdir(person_path):
                img_path = os.path.join(person_path, img_name)
                img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)
                if img is not None and img.shape[0] > 0 and img.shape[1] > 0:
                    images.append(img)
                    labels.append(current_label)
                else:
                    print(f"Failed to load or invalid image: {img_path}")
            current_label += 1
    return images, labels, label_dict
dataset_path = "D:/23108106/Dataset/Faces"
images, labels, label_dict = load_dataset(dataset_path)
recognizer = cv2.face.LBPHFaceRecognizer_create()
recognizer.train(images, np.array(labels))
recognizer.save('face_recognizer.yml')
with open("label_map.txt", "w") as f:
    for label, name in label_dict.items():
        f.write(f"{label}:{name}\n")
recognizer.read("face_recognizer.yml")
label_map = {}
with open("label_map.txt", "r") as f:
    for line in f:
        label, name = line.strip().split(":")
        label_map[int(label)] = name
face_cascade = cv2.CascadeClassifier(cv2.data.haarcascades + 'haarcascade_frontalface_default.xml')
test_img_path = "D:/23108106/A23.jpg"
```

```python
img = cv2.imread(test_img_path)
if img is None:
    raise ValueError("Test image not loaded! Check the path.")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
faces = face_cascade.detectMultiScale(gray, scaleFactor=1.1, minNeighbors=3, minSize=(30, 30))
confidence_threshold = 80
if len(faces) == 0:
    print("No faces detected!")
else:
    for (x, y, w, h) in faces:
        roi_gray = gray[y:y+h, x:x+w]
        label, confidence = recognizer.predict(roi_gray)
        name = label_map.get(label, "Unknown")
        if confidence < confidence_threshold:
            display_name = name
            print(f"Unlocked: {name} (Confidence: {int(confidence)})")
        else:
            display_name = "Unknown"
            print(f"Access Denied (Confidence: {int(confidence)})")
        cv2.rectangle(img, (x, y), (x+w, y+h), (0, 255, 0), 2)
        cv2.putText(img, f"{display_name} ({int(confidence)})", (x, y-10),
                    cv2.FONT_HERSHEY_SIMPLEX, 0.8, (0, 255, 0), 2)
plt.figure(figsize=(8, 5))
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.axis('off')
plt.show()
```

```
Access Denied (Confidence: 84)
```



Case Study 1: Smart Attendance System for Universities

A university wants to automate its classroom attendance system using facial recognition to avoid manual entry and prevent proxy attendance.

Task: Build a system that:

- **Detects faces from a live camera feed.**
- **Recognizes and identifies students using a trained face recognition model.**
- **Marks their attendance by logging names and timestamps.**

Application: Automated attendance in classrooms, labs, or exams.

```python
# Case Study 1: Smart Attendance System for Universities
import cv2
import numpy as np
import os
import pandas as pd
from datetime import datetime
from matplotlib import pyplot as plt
from IPython.display import clear_output
def load_dataset(dataset_path):
    images, labels, label_map = [], [], {}
    current_label = 0
    for person in os.listdir(dataset_path):
        person_folder = os.path.join(dataset_path, person)
        if os.path.isdir(person_folder):
            label_map[current_label] = person
            for img_name in os.listdir(person_folder):
                img_path = os.path.join(person_folder, img_name)
                img = cv2.imread(img_path)
                if img is None:
                    continue
                gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
                images.append(gray)
                labels.append(current_label)
            current_label += 1
    return images, labels, label_map
dataset_path = "D:/23108106/Dataset/Faces"
images, labels, label_map = load_dataset(dataset_path)
recognizer = cv2.face.LBPHFaceRecognizer_create()
recognizer.train(images, np.array(labels))
attendance = {}
face_cascade = cv2.CascadeClassifier(cv2.data.haarcascades + "haarcascade_frontalface_default.xml")
cap = cv2.VideoCapture("D:/23108106/Lecture A.mp4")
frame_skip = 5
frame_count = 0
while True:
    ret, frame = cap.read()
    if not ret:
        break
    frame_count += 1
    if frame_count % frame_skip != 0:
```

```
        continue
    h, w = frame.shape[:2]
    scale = 640 / w
    small_frame = cv2.resize(frame, (640, int(h * scale)))
    gray = cv2.cvtColor(small_frame, cv2.COLOR_BGR2GRAY)
    faces = face_cascade.detectMultiScale(gray, scaleFactor=1.2, minNeighbors=4)
    for (x, y, w_f, h_f) in faces:
        roi_gray = gray[y:y+h_f, x:x+w_f]
        label, conf = recognizer.predict(roi_gray)
        name = label_map.get(label, "Unknown")
        if conf < 80:
            if name not in attendance:
                attendance[name] = datetime.now().strftime("%H:%M:%S")
        x1, y1 = int(x / scale), int(y / scale)
        x2, y2 = int((x + w_f) / scale), int((y + h_f) / scale)
        cv2.rectangle(frame, (x1, y1), (x2, y2), (0, 255, 0), 2)
        cv2.putText(frame, f"{name}", (x1, y1 - 10),
                    cv2.FONT_HERSHEY_SIMPLEX, 0.8, (0, 255, 0), 2)
    plt.figure(figsize=(10,6))
    plt.imshow(cv2.cvtColor(frame, cv2.COLOR_BGR2RGB))
    plt.axis('off')
    clear_output(wait=True)
    plt.show()
cap.release()
df = pd.DataFrame(list(attendance.items()), columns=["Name", "Time"])
df.to_csv("attendance_log.csv", index=False)
print("Attendance saved -> attendance_log.csv")
```

Case Study 2: Access Control in Restricted Zones

A research lab requires a secure entry system where only authorized personnel can access sensitive rooms. Face recognition is used in place of keycards or biometric fingerprints.

Task: Develop a system that:

- **Captures a face at the door entry.**
- **Verifies identity using a stored face database.**
- **Allows or denies access based on recognition.**

Application: Security in labs, data centers, or military zones.

```python
# Case Study 2: Access Control in Restricted Zones
dataset_path = "D:/23108106/Dataset/Faces"
images, labels, label_map = [], [], {}
current_label = 0
for person in os.listdir(dataset_path):
    person_folder = os.path.join(dataset_path, person)
    if os.path.isdir(person_folder):
        label_map[current_label] = person
        for img in os.listdir(person_folder):
            img_path = os.path.join(person_folder, img)
            img = cv2.imread(img_path)
            if img is None: continue
            gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
            images.append(gray)
            labels.append(current_label)
        current_label += 1
recognizer = cv2.face.LBPHFaceRecognizer_create()
recognizer.train(images, np.array(labels))
cap = cv2.VideoCapture("D:/23108106/Lecture A.mp4")
face_cascade = cv2.CascadeClassifier(cv2.data.haarcascades + "haarcascade_frontalface_default.xml")
frame_skip = 5
frame_count = 0
while True:
    ret, frame = cap.read()
    if not ret: break
    frame_count += 1
    if frame_count % frame_skip != 0: continue
    h, w = frame.shape[:2]
    scale = 640/w
    small_frame = cv2.resize(frame, (640, int(h*scale)))
    gray = cv2.cvtColor(small_frame, cv2.COLOR_BGR2GRAY)
    faces = face_cascade.detectMultiScale(gray, 1.2, 4)
    for (x, y, w_f, h_f) in faces:
        roi_gray = gray[y:y+h_f, x:x+w_f]
        label, conf = recognizer.predict(roi_gray)
        status = "Access Granted" if conf < 80 else "Access Denied"
        color = (0,255,0) if conf < 80 else (0,0,255)
        name = label_map.get(label, "Unknown") if conf < 80 else "Unknown"
        x1, y1 = int(x/scale), int(y/scale)
        x2, y2 = int((x+w_f)/scale), int((y+h_f)/scale)
```

```
        name = label_map.get(label, "Unknown") if conf < 80 else "Unknown"
        x1, y1 = int(x/scale), int(y/scale)
        x2, y2 = int((x+w_f)/scale), int((y+h_f)/scale)
        cv2.rectangle(frame, (x1, y1), (x2, y2), color, 2)
        cv2.putText(frame, f"{name} - {status}", (x1, y1-10),
                    cv2.FONT_HERSHEY_SIMPLEX, 0.7, color, 2)
    plt.figure(figsize=(10,6))
    plt.imshow(cv2.cvtColor(frame, cv2.COLOR_BGR2RGB))
    plt.axis('off')
    clear_output(wait=True)
    plt.show()

cap.release()
```

Case Study 3: Face-Based Tagging in Photo Archives

A digital media company wants to automatically tag celebrities in their archives for better organization and search ability.

Task: Create a face recognition solution that:

- **Trains on a dataset of known public figures.**
- **Automatically labels new incoming images with the names of recognized individuals.**

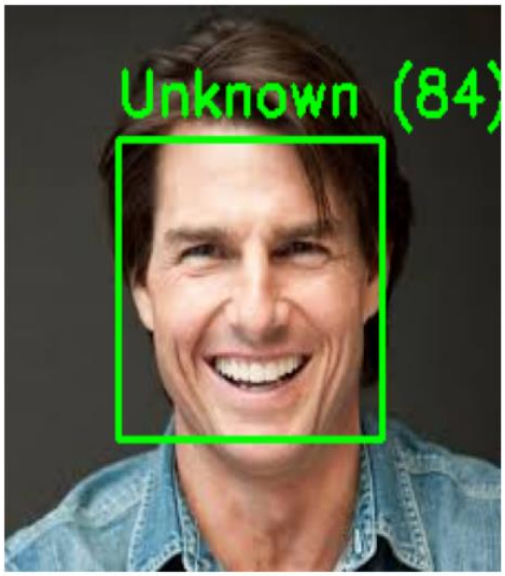
Application: Media tagging, celebrity tracking, social media filters.

```python
# Case Study 3: Face-Based Tagging in Photo Archives
import os
import cv2
import numpy as np
from matplotlib import pyplot as plt
def load_dataset(dataset_path):
    images = []
    labels = []
    label_dict = {}
    current_label = 0
    for person_name in os.listdir(dataset_path):
        person_path = os.path.join(dataset_path, person_name)
        if os.path.isdir(person_path):
            label_dict[current_label] = person_name
            for img_name in os.listdir(person_path):
                img_path = os.path.join(person_path, img_name)
                img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)
                if img is not None and img.shape[0] > 0 and img.shape[1] > 0:
                    images.append(img)
                    labels.append(current_label)
                else:
                    print(f"Failed to load or invalid image: {img_path}")
            current_label += 1
    return images, labels, label_dict
dataset_path = "D:/23108106/Dataset/Faces"
images, labels, label_dict = load_dataset(dataset_path)
recognizer = cv2.face.LBPHFaceRecognizer_create()
recognizer.train(images, np.array(labels))
recognizer.save('D:/23108106/face_recognizer.yml')
with open("label_map.txt", "w") as f:
    for label, name in label_dict.items():
        f.write(f"{label}:{name}\n")
recognizer.read('D:/23108106/face_recognizer.yml')
label_map = {}
with open("label_map.txt", "r") as f:
    for line in f:
        label, name = line.strip().split(":")
        label_map[int(label)] = name
face_cascade = cv2.CascadeClassifier(cv2.data.haarcascades + 'haarcascade_frontalface_default.xml')
test_img_path = "D:/23108106/A23.jpg"
```

```python
img = cv2.imread(test_img_path)
if img is None:
    raise ValueError("Test image not loaded! Check the path.")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
faces = face_cascade.detectMultiScale(
    gray,
    scaleFactor=1.1,
    minNeighbors=3,
    minSize=(30, 30)
)
confidence_threshold = 80
if len(faces) == 0:
    print("No faces detected!")
else:
    for (x, y, w, h) in faces:
        roi_gray = gray[y:y+h, x:x+w]
        label, confidence = recognizer.predict(roi_gray)
        name = label_map.get(label, "Unknown")
        display_name = name if confidence < confidence_threshold else "Unknown"
        cv2.rectangle(img, (x, y), (x+w, y+h), (0, 255, 0), 2)
        cv2.putText(img, f"{display_name} ({int(confidence)})",
                    (x, y-10), cv2.FONT_HERSHEY_SIMPLEX, 0.8, (0, 255, 0), 2)
        if display_name != "Unknown":
            print(f"Unlocked: {display_name} (Confidence: {int(confidence)})")
        else:
            print(f"Access Denied (Confidence: {int(confidence)})")
```

```python
plt.figure(figsize=(8,5))
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.axis('off')
plt.show()
```

```
Access Denied (Confidence: 84)
```



Case Study 4: Real-Time Intruder Detection Using Face Detection
A home surveillance system uses face detection to monitor activity. If a human face is detected at an unexpected time (e.g., late night), the system raises an alert.
Task: Design a face detection system that:

- **Continuously monitors camera feeds.**
- **Detects and logs any human face.**
- **Optionally sends alerts when unfamiliar faces are detected or during restricted hours.**

Application: Smart home security and intrusion detection.

```
# Case Study 4: Real-Time Intruder Detection Using Face Detection
video_path = "D:/23108106/SSA.mp4"
cap = cv2.VideoCapture(video_path)
frame_skip = 5
frame_count = 0
while True:
    ret, frame = cap.read()
    if not ret: break
    frame_count += 1
    if frame_count % frame_skip != 0: continue
    h, w = frame.shape[:2]
    scale = 640/w
    small_frame = cv2.resize(frame, (640, int(h*scale)))
    gray = cv2.cvtColor(small_frame, cv2.COLOR_BGR2GRAY)
    faces = face_cascade.detectMultiScale(gray, 1.2, 4)
    for (x,y,w_f,h_f) in faces:
        roi_gray = gray[y:y+h_f, x:x+w_f]
        label, conf = recognizer.predict(roi_gray)
        if conf >= 80:
            x1, y1 = int(x/scale), int(y/scale)
            x2, y2 = int((x+w_f)/scale), int((y+h_f)/scale)
            cv2.rectangle(frame, (x1,y1), (x2,y2), (0,0,255), 2)
            cv2.putText(frame, "Intruder", (x1,y1-10), cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0,0,255), 2)
            print("Intruder detected!")
```

```
        print('Intruder detected')
    plt.figure(figsize=(10,6))
    plt.imshow(cv2.cvtColor(frame, cv2.COLOR_BGR2RGB))
    plt.axis('off')
    clear_output(wait=True)
    plt.show()
cap.release()
```

Case Study 5: Criminal Identification Using Face Recognition in Law Enforcement
Law enforcement agencies want to identify known criminals in public areas using CCTV footage and a criminal face database.
Task: Build a system that:

- **Detects and recognizes faces in real-time from surveillance cameras.**
- **Matches against a database of wanted individuals.**
- **Sends alerts when a match is found.**

Application: Criminal tracking, missing person identification, border control.

```
# Case Study 5: Criminal Identification Using Face Recognition in Law Enforcement
import cv2
import os
import numpy as np
from matplotlib import pyplot as plt
from IPython.display import clear_output
dataset_path = "D:/23108106/Dataset/Faces"
images, labels, label_map = [], [], {}
current_label = 0
for criminal in os.listdir(dataset_path):
    folder = os.path.join(dataset_path, criminal)
    if os.path.isdir(folder):
        label_map[current_label] = criminal
        for img_name in os.listdir(folder):
            img_path = os.path.join(folder, img_name)
            img = cv2.imread(img_path)
            if img is None: continue
            gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
            images.append(gray)
            labels.append(current_label)
        current_label += 1
recognizer = cv2.face.LBPHFaceRecognizer_create()
recognizer.train(images, np.array(labels))
video_path = "D:/23108106/SSA.mp4"
cap = cv2.VideoCapture(video_path)
face_cascade = cv2.CascadeClassifier(cv2.data.haarcascades + "haarcascade_frontalface_default.xml")
frame_skip = 6
frame_count = 0
detected_criminals = {}
while True:
    ret, frame = cap.read()
    if not ret:
        break
    frame_count += 1
    if frame_count % frame_skip != 0:
        continue
    h, w = frame.shape[:2]
    scale = 640 / w
    small_frame = cv2.resize(frame, (640, int(h * scale)))
    gray = cv2.cvtColor(small_frame, cv2.COLOR_BGR2GRAY)
```

```
    faces = face_cascade.detectMultiScale(gray, 1.2, 4)
    for (x, y, w_f, h_f) in faces:
        roi_gray = gray[y:y + h_f, x:x + w_f]
        label, conf = recognizer.predict(roi_gray)
        if conf < 80:
            name = label_map[label]
            detected_criminals[name] = detected_criminals.get(name, 0) + 1
            color = (0, 0, 255)  # Red for criminal
            status_text = f"Criminal: {name} ({int(conf)})"
            print(f"Alert! Criminal Detected: {name} with confidence {int(conf)}")
        else:
            name = "Unknown"
            color = (0, 255, 0)
            status_text = "Unknown"
        x1, y1 = int(x / scale), int(y / scale)
        x2, y2 = int((x + w_f) / scale), int((y + h_f) / scale)
        cv2.rectangle(frame, (x1, y1), (x2, y2), color, 2)
        cv2.putText(frame, status_text, (x1, y1 - 10),
                    cv2.FONT_HERSHEY_SIMPLEX, 0.7, color, 2)
```

```
    plt.figure(figsize=(10, 6))
    plt.imshow(cv2.cvtColor(frame, cv2.COLOR_BGR2RGB))
    plt.axis('off')
    clear_output(wait=True)
    plt.show()
cap.release()
import pandas as pd
df = pd.DataFrame(list(detected_criminals.items()), columns=["Name", "Times_Detected"])
df.to_csv("detected_criminals.csv", index=False)
print("Detection log saved -> detected_criminals.csv")
```

```
Detection log saved -> detected_criminals.csv
```

https://github.com/Ayesha-Munir-Chohan/